

[18]

SARDAR PATEL UNIVERSITY
M.Sc. (CA & IT) EXAMINATION, IX SEM

Date : 29th Dec 2020, Tuesday
Session : Morning
Course No : PS09CIIT02

Time : 10:00am To 12:00pm
Sub: Data Warehouse and Data Mining
Total marks : 70

Q – 1 Multiple Choice Question

[10]

- i) Which one manages both current and historic transactions?
a) OLTP b) OLAP c) Spread Sheet d) All of above
- ii) The allows the selection of the relevant information necessary for the data warehouse
a) top-down view b) data warehouse view
c) data source view d) business query view
- iii) _____ is the specialized data warehouse database.
a) Oracle b) DBZ c) Informix d) Redbrick
- iv) If T consist of 500000 transactions, 20000 transaction contain bread, 30000 transaction contain jam, 10000 transaction contain both bread and jam. Then the confidence of buying bread with jam is _____.
a) 33.33%. b) 66.66%. c) 45% d) 50%
- v) _____ is used to proceed from very specific knowledge to more general information.
a) Induction b) Compression c) Approximation d) Substitution
- vi) The second phase of A Priori algorithm is _____.
a) Candidate generation. b) Item set generation. c) Pruning d) Partitioning
- vii) The absolute number of transactions supporting X in T is called _____.
a) confidence. b) support. c) Support Count d) None
- viii) _____ data consists of sample input data as well as the classification assignment for the data.
a) Missing. b) Measuring. c) Non-training. d) Training.
- ix) Each neuron is made up of a number of nerve fibres called _____.
a) electrons. b) molecules. c) atoms. d) dendrites.
- x) The sigmoid function also known as _____ functions.
a) Regression b) Logistic c) Probability d) Neural

Q – 2 Fill in the Blanks and True – False

[08]

- i) Data warehouse database server is the heart of warehouse. (True / False)
- ii) _____ describes the data contained in the data warehouse.
- iii) Link Analysis is otherwise called as affinity analysis. (True / False)
- iv) Treating incorrect or missing data is called as _____.
- v) The value says that transactions in D that support X also support Y is called confidence. (True / False)
- vi) The right hand side of an association rule is called _____.
- vii) Non – Training data consists of sample input data as well as the classification assignment for the data. (True / False)
- viii) Rule based classification algorithms generate _____ rule to perform the classification.

Q – 3 Short Answer **attempt any ten (Each carry 2 marks)**

[20]

- i) Write difference between Data Warehouse and Data Mart
- ii) Draw a figure of data warehouse building blocks or components
- iii) Write significant factors which drove companies for Data Warehousing
- iv) Explain Types of Metadata.
- v) Explain cluster analysis
- vi) Write major task in data processing
- vii) Write about market basket analysis
- viii) Explain improving the efficiency of Apriori.
- ix) What do you mean by Classification
- x) Write about data classification process
- xi) What is Bayesian belief network
- xii) What do you mean by classification by Backpropagation

Q – 4 Attempt any four (**Each carry 8 marks**)

[32]

- i) Explain STAR Schema in detail
- ii) Explain ROLAP versus MOLAP.
- iii) Explain Data Integration in detail
- iv) Explain Data Cleaning in detail
- v) Which Patterns are interesting? Discuss pattern evaluation methods
- vi) Write Mining Closed and Max Pattern.
- vii) Explain Decision Tree induction
- viii) Discuss Back Propagation.

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